

Finite Spectral-Gap Certificates in a Riemann Prime-Resolvent Programme

Lean verification, a subsequential criterion, and a falsification campaign

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<https://github.com/lluiseiriksson/riemann-prime-resolvent>

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Scope statement. This article does not prove the Riemann hypothesis. It proves finite Hilbert-space and spectral-gap estimates, reports their Lean 4 formalization, and records a conditional analytic route whose concrete asymptotic inputs remain open.

Abstract

We isolate a finite-dimensional estimate used in a prime-resolvent research programme related to the Riemann Ξ -function. Let T be a real symmetric operator, let g be a normalized ground eigenvector with eigenvalue λ_0 , and let u be a normalized trial vector. If the ground state is separated from the rest of the spectrum by $\Delta > 0$, then

$$\Delta(1 - \langle g, u \rangle^2) \leq \langle Tu, u \rangle - \lambda_0, \quad \|g - u\| \leq \sqrt{\frac{2(\langle Tu, u \rangle - \lambda_0)}{\Delta}}$$

after sign alignment. Returning from a normalized Galerkin projection to a complete trial vector costs at most twice the projection tail. We formalize these statements in Lean 4, including a guard proving that a positive indexed gap forces a one-dimensional ground eigenspace. Since Mathlib orders the finite eigenvalue list decreasingly, a single inequality between the final two entries suffices to certify the universal gap.

We then state a repaired subsequential criterion: along a cofinal family of finite approximants, a defect

$$\mathfrak{d}_j = \|k_j\|_2 \sqrt{2r_j/\Delta_j} + 2 \text{tail}_j$$

with suitable polynomial decay, together with finite parity, non-vanishing and local analytic convergence hypotheses, would imply the Riemann hypothesis. Those hypotheses are not established here. Finally, we report a registered finite falsification campaign. Its global verdict was INCONCLUSIVE; an independent polynomial-root witness diagnosed one missing scan root at $(\lambda, N) = (3, 40)$ as a base-resolution failure. The numerical experiment is not used in any proof.

Keywords: Riemann hypothesis; spectral gap; Rayleigh quotient; Galerkin approximation; Stieltjes transform; Hausdorff moments; formal verification; Lean 4.

1 Introduction and claim boundary

Write

$$\Xi(z) = \xi\left(\frac{1}{2} + iz\right), \quad \mathcal{S}_\Xi(x) = \frac{1}{2\sqrt{x}} \frac{\xi'}{\xi}\left(\frac{1}{2} + \sqrt{x}\right), \quad x > 0,$$

where ξ is the completed Riemann zeta function. A companion criterion packages the real-zero question for Ξ as a one-point Hausdorff moment problem. The construction side of the programme asks whether positive finite spectral approximants can converge non-circularly to $\mathcal{S}_\Xi$.

The purpose of this paper is narrower. It closes a finite spectral estimate that had previously appeared only as an informal obligation. It also records exactly what the estimate does not provide. In particular, it does not construct the required infinite operator family, prove the necessary prime-archimedean cancellation, or establish convergence to Ξ .

The contributions are:

- (i) an elementary finite spectral-gap inequality and its Galerkin-tail completion;
- (ii) a formal degeneracy guard: the positive indexed gap implies that the ground eigenspace has dimension one;
- (iii) a reduction of the universal gap certificate to the last adjacent gap in Mathlib's decreasingly ordered eigenvalue list;
- (iv) a Lean 4 implementation checked without project-specific axioms or placeholders;
- (v) a precise statement of a conditional subsequential route and a separately labelled numerical falsification campaign.

Classical background on ζ and ξ may be found in [1, 2]. The Hausdorff moment theorem is due to Hausdorff [3]; standard references include [4, 5]. The finite spectral estimates used below are in the family of standard Rayleigh, Temple and Kato arguments [6].

2 The one-point criterion and the open construction arrow

Fix $x_0 > 0$ and define the normalized derivative jet

$$b_n(x_0) = x_0^n \frac{(-1)^n}{n!} \mathcal{S}_\Xi^{(n)}(x_0).$$

For a sequence b , define $D^0 b_n = b_n$ and

$$D^{k+1} b_n = D^k b_n - D^k b_{n+1}.$$

The documented analytic argument in the companion criterion layer has the following form.

Proposition 2.1 (Documented one-point criterion). *For a fixed $x_0 > 0$, subject to the stated branch and logarithmic-derivative conventions, the following are equivalent:*

- (a) *all nontrivial zeros of ζ lie on the critical line;*
- (b) *$(b_n(x_0))_{n \geq 0}$ is a Hausdorff moment sequence on $[0, 1]$;*
- (c) *$D^k b_n(x_0) \geq 0$ for all $n, k \geq 0$.*

Under the Riemann hypothesis the representing atoms are obtained from the ordinates γ_j by

$$v_j = \frac{x_0}{x_0 + \gamma_j^2} \in (0, 1), \quad b_n(x_0) = \sum_j \frac{1}{x_0 + \gamma_j^2} v_j^n.$$

Conversely, a representing measure yields a Stieltjes transform holomorphic off the negative real axis and agreeing locally with $\mathcal{S}_\Xi$.

Proposition 2.1 is a documented analytic theorem, not a fully closed Lean theorem in the present repository. The difficult construction arrow is

$$\text{prime-built finite spectra} \longrightarrow \text{positive Stieltjes limit agreeing with } \mathcal{S}_\Xi.$$

The finite result below addresses one estimate inside that arrow.

3 Finite spectral-gap control

Let E be a finite-dimensional real Hilbert space. Let $T : E \rightarrow E$ be symmetric and choose an orthonormal eigenbasis $(e_i)_{i \in I}$ with eigenvalues $(\lambda_i)_{i \in I}$. Fix an index i_0 and write

$$g = e_{i_0}, \quad \lambda_0 = \lambda_{i_0}.$$

Let $u \in E$ satisfy $\|u\| = 1$ and put

$$c_i = \langle e_i, u \rangle, \quad r = \langle Tu, u \rangle - \lambda_0.$$

Theorem 3.1 (Finite Rayleigh-gap inequality). *Suppose that for some $\Delta \in \mathbb{R}$,*

$$\lambda_0 + \Delta \leq \lambda_i \quad \text{for every } i \neq i_0.$$

Then

$$\Delta(1 - \langle g, u \rangle^2) \leq r.$$

Proof. Parseval's identity and diagonalization give

$$\sum_i c_i^2 = 1, \quad \langle Tu, u \rangle = \sum_i \lambda_i c_i^2.$$

Consequently

$$\begin{aligned} r &= \sum_i (\lambda_i - \lambda_0) c_i^2 \\ &= \sum_{i \neq i_0} (\lambda_i - \lambda_0) c_i^2 \\ &\geq \Delta \sum_{i \neq i_0} c_i^2 = \Delta(1 - c_{i_0}^2). \end{aligned}$$

□

Corollary 3.2 (Distance bound). *Assume the hypotheses of Theorem 3.1, $\Delta > 0$, and choose the sign of g so that $\langle g, u \rangle \geq 0$. Then $r \geq 0$ and*

$$\|g - u\| \leq \sqrt{\frac{2r}{\Delta}}.$$

Proof. For $c = \langle g, u \rangle \in [0, 1]$,

$$\|g - u\|^2 = 2(1 - c) \leq 2(1 - c^2) \leq \frac{2r}{\Delta}.$$

□

3.1 Why the gap hypothesis encodes simplicity

The strict positivity of Δ is essential. If the ground eigenvalue were degenerate, a normalized u could lie in the same eigenspace while being orthogonal to a chosen g . Then $r = 0$ but $\|g - u\| = \sqrt{2}$, contradicting the claimed distance estimate.

The indexed gap rules out this counterexample because an eigenbasis enumerates multiplicity.

Corollary 3.3 (Degeneracy guard). *If $\Delta > 0$ and $\lambda_0 + \Delta \leq \lambda_i$ for every $i \neq i_0$, then i_0 is the unique index with eigenvalue λ_0 . For a symmetric operator this implies*

$$\dim \ker(T - \lambda_0 I) = 1.$$

In the formal proof the final statement is not inferred from informal language. Mathlib identifies the cardinality of the filtered eigenvalue list with the finite dimension of the eigenspace. The positive gap makes that filtered list the singleton $\{i_0\}$.

3.2 One adjacent inequality is sufficient

Mathlib's finite spectral theorem orders the eigenvalues of a symmetric operator decreasingly. Thus, for a list

$$\lambda_0^\downarrow \geq \lambda_1^\downarrow \geq \dots \geq \lambda_{m-2}^\downarrow \geq \lambda_{m-1}^\downarrow,$$

the ground eigenvalue is the last entry.

Theorem 3.4 (Adjacent-gap reduction). *Let $m \geq 2$ and suppose the finite eigenvalue list is antitone. If*

$$\lambda_{m-1}^\downarrow + \Delta \leq \lambda_{m-2}^\downarrow,$$

then for every $i \neq m - 1$,

$$\lambda_{m-1}^\downarrow + \Delta \leq \lambda_i^\downarrow.$$

If $\Delta > 0$, the last eigenvalue is simple.

Proof. Every index other than the last is at most the penultimate index. Antitonicity therefore gives $\lambda_{m-2}^\downarrow \leq \lambda_i^\downarrow$, and the result follows by transitivity. Simplicity follows from Corollary 3.3. $\square$

This reduction matters computationally: an interval certificate for the two lowest eigenvalues can supply the entire indexed-gap premise.

4 Galerkin projection and the exact defect

Let k be a complete trial vector and $P_N k$ its finite Galerkin projection. Assume $P_N k \neq 0$ and set

$$u = \frac{P_N k}{\|P_N k\|}, \quad \text{tail} = \|k - P_N k\|.$$

Let g_N be the aligned normalized ground eigenvector of the finite symmetric operator, let $\Delta_N > 0$ be its spectral gap, and let

$$r_N = \langle T_N u, u \rangle - \lambda_{0,N}.$$

Theorem 4.1 (Finite Galerkin defect). *Under the preceding assumptions,*

$$\| \|k\| g_N - k \| \leq \|k\| \sqrt{\frac{2r_N}{\Delta_N}} + 2 \text{tail}.$$

The same positive adjacent-gap certificate proves simultaneously that the finite ground eigenspace is one-dimensional.

Proof. Corollary 3.2 gives

$$\|g_N - u\| \leq \sqrt{2r_N/\Delta_N}.$$

Insert $\|k\| u$, $P_N k = \|P_N k\| u$, and k into a triangle inequality:

$$\begin{aligned} \| \|k\| g_N - k \| &\leq \|k\| \|g_N - u\| + \| \|k\| - \|P_N k\| \| + \|P_N k - k\| \\ &\leq \|k\| \sqrt{\frac{2r_N}{\Delta_N}} + 2 \|k - P_N k\|. \end{aligned}$$

The second inequality uses the reverse triangle inequality. No small-tail assumption such as $\text{tail} \leq \|k\|/2$ is needed. $\square$

We therefore define the finite defect

$$\mathfrak{d}_N = \|k\| \sqrt{\frac{2r_N}{\Delta_N}} + 2 \text{tail}. \quad (1)$$

The factor 2 is not decorative: one tail term changes the scale $\|P_N k\|$ to $\|k\|$ and the other returns from $P_N k$ to k .

5 A conditional subsequential route

The finite inequality is useful only if it is coupled to a concrete family. We record the current conditional route because it identifies the exact asymptotic obligation rather than hiding it in a generic convergence premise.

Let (λ_j, N_j) be a cofinal diagonal of finite approximants. For the j th projected normalized trial vector define r_j , Δ_j and tail_j as above, and put

$$\mathfrak{d}_j = \|k_j\|_2 \sqrt{\frac{2r_j}{\Delta_j}} + 2 \text{tail}_j.$$

Proposition 5.1 (Conditional RATE_Σ criterion). *Suppose there are $0 < a_1 < a_2 < a_3 < 1/2$ such that:*

- (P1) *along the cofinal diagonal, the relevant finite ground states are simple, have the required parity, and satisfy the non-vanishing condition needed by the finite zero-identification theorem;*
- (P2) *the transforms $\widehat{k}_j$ converge uniformly to Ξ on a compact two-dimensional neighborhood of the segment $i[a_1, a_2]$ contained in $|\text{Im } z| \leq a_3$;*
- (P3) *the same approximants satisfy*

$$\lambda_j^{a_3} \sqrt{\log \lambda_j} \mathfrak{d}_j \longrightarrow 0.$$

Then the documented Stieltjes and identity-theorem argument implies the Riemann hypothesis.

This proposition is not presently a kernel-checked end-to-end theorem. The finite estimate (1) is formalized, but the parity and non-vanishing family, the local convergence to Ξ , the infinite atomic compactness instantiation, and the decay in (P3) remain open. In particular, the proposition must not be quoted as an unconditional theorem about a concrete operator.

The subsequential form reduces the verification surface: it uses finite real zeros directly and requires decay only on a cofinal diagonal. It does not remove the arithmetic content. Any positive power decay in (P3) still requires signed cancellation not supplied by positive majorants alone.

6 Lean 4 formalization

The results of Sections 3 and 4 are implemented in the repository’s `SpectralDefect.lean` module. The implementation uses Mathlib’s finite spectral theorem, orthonormal eigenvector basis, Parseval identity and eigenvalue multiplicity count. Lean and Mathlib are described in [7, 8].

The main declaration chain is:

Declaration	Mathematical content
<code>spectralGap_mul_one_sub_inner_sq_le_rayleighExcess_of_eigenbasis</code>	Theorem 3.1 in an arbitrary orthonormal eigenbasis.
<code>spectralGap_mul_one_sub_inner_sq_le_rayleighExcess_of_isSymmetric</code>	Instantiation with Mathlib’s finite spectral theorem.
<code>finrank_groundEigenspace_eq_one_of_positive_gap</code>	The ground eigenspace has dimension one.
<code>antitone_gap_from_penultimate</code>	The adjacent-gap reduction of Theorem 3.4.
<code>norm_scaled_eigenvectorBasis_sub_complete_le_rayleighGapDefect</code>	The finite Galerkin estimate (1).
<code>simpleGround_and_norm_scaled_eigenvectorBasis_sub_complete_le_of_adjacentGap</code>	Combined adjacent-gap, simplicity and Galerkin certificate.

The project oracle prints the axioms of each public theorem. For the declarations above it reports only the standard logical dependencies `propext`, `Classical.choice` and `Quot.sound`. The checked source contains no `sorry`, `admit` or project-specific axiom.

This verification claim concerns the finite theorems in this article. It does not certify Proposition 2.1, the conditional RATE_Σ criterion as an end-to-end result, or the missing concrete operator construction.

7 Registered finite falsification campaign

A separate campaign tested a concrete finite spectral arm. Its role was falsification: predetermined gates could kill that arm at the probed scales, but no finite outcome could prove the Riemann hypothesis or the asymptotic criterion.

The registered grid was

$$\lambda \in \{2, \sqrt{8}, 3, \sqrt{13}\}, \quad N \in \{20, 40, 80, 120\},$$

with a base and a high-resolution scan for each of the sixteen grid cells. The global output was

```
T3-VERDICT: INCONCLUSIVE conv=S-half;x0=1;
grid=L{2,sqrt8,3,sqrt13}xN{20,40,80,120}
detail={"invalid": ["3:40", "3:80", "3:120"], "missing": []}
```

The three invalid entries were base scans at $\lambda = 3$ with a deficit of one detected positive root. High-resolution replicas were valid. Across the thirteen valid cells, the registered kill observables reported no low-energy spurious modes and zero low-energy mass under the specified threshold. This is a non-kill result at finite scales, not evidence of the required limit.

7.1 Polynomial completeness witness at $(3, 40)$

The A4 witness replaced sign-change scanning by the characteristic polynomial of the finite problem. For $(\lambda, N) = (3, 40)$, the reported certificate had degree 80 and found 80 real roots, split into 40 positive and 40 negative roots, with $P(0) \neq 0$. The root missed by the base scan was

$$z = 82.9103880394\dots,$$

whose nearest neighbor was approximately 1.8251 away. The high-resolution scan found it; the base scan did not. The mechanical A4 classification was BASE-RESOLUTION, not a de-duplication merge, window failure or even multiplicity.

This diagnosis is certified for $(3, 40)$. The matching deficit pattern at $(3, 80)$ and $(3, 120)$ is suggestive but is not promoted here to a certified cause without separate completeness witnesses. Moreover, A4 was explicitly non-gating: it did not alter the registered INCONCLUSIVE verdict.

8 What remains open

The finite theorem removes one informal step but leaves the decisive mathematics intact. A complete argument would still require:

- (1) a concrete Hilbert space and operator family with domains and self-adjointness proved non-circularly;
- (2) parity, non-vanishing and adjacent positive-gap certificates along an infinite cofinal diagonal;
- (3) certified control of r_j , Δ_j and tail_j strong enough for RATE_Σ ;
- (4) convergence of the transforms of the same trial vectors to Ξ on the required two-dimensional compact set;
- (5) completion of the infinite-measure Stieltjes compactness and one-point criterion in Lean;
- (6) independent mathematical review of the analytic bridges and all normalization conventions.

Numerical agreement, a finite simple spectrum, or a green Lean build cannot replace these obligations. The present result should therefore be read as a verified finite lemma and a sharper research interface.

9 Conclusion

We proved and formalized the finite estimate

$$\| \|k\| g_N - k \| \leq \|k\| \sqrt{2r_N/\Delta_N} + 2\text{tail}_N.$$

The formal statement prevents a common degeneracy error: its positive gap forces the ground eigenspace to have dimension one. Because the finite eigenvalue list is antitone, the full gap premise reduces to one certified inequality between the last two eigenvalues. This is a genuine simplification of the finite verification task.

It is not a proof of the Riemann hypothesis. The asymptotic same-approximant convergence and cancellation problem remains the central open step.

Software, data and reproducibility

The formal source audited for this version is the public commit [4b986b65232e](#) on branch `galerkin/rayleigh-gap-bound` of [riemann-prime-resolvent](#). That commit contains the Lean diff, this $\text{T}_{\text{E}}\text{X}$ source and the version 1.0 PDF in one tree. The present version 1.1 adds this reproducibility record; its archival release should again place source and rendered PDF in one commit.

The checked environment was Lean 4.31.0 (commit [68218e876d2a](#)) with Lake 5.0.0 and Mathlib revision [fabf563a7c95](#). On 6 August 2026 the following commands completed with exit code zero:

```
lake build RiemannPrimeResolvent.SpectralDefect [8558/8558]
lake build [8567/8567]
lake env lean oracle_check.lean
```

For the two new declarations, the oracle reported only `propext`, `Classical.choice` and `Quot.sound`; it reported no `sorryAx`. A plain-text verification capsule distributed beside the paper records the commands, revisions and output hashes.

The campaign record is the public commit [382e1ab4aaec](#) on branch `campaign/t3-kill-test` of [riemann-resolvent-programme](#); it is also the head of pull request 47 at the time of audit. Its registered-verdict file states an important provenance limitation: the table and witnesses are a verbatim screen transcription from the Colab runtime. The attempted zip of the 32 raw checkpoints did not reach the desktop and is not an archived artifact in that commit. The probe is deterministic and the checkpoints are reported reproducible, but this paper does not upgrade the transcription to independent raw-data custody.

AI-assistance disclosure

The author directed an AI-assisted drafting and formalization workflow. Mathematical claims included as Lean theorems were checked by the Lean kernel; the author remains responsible for the manuscript, its submission, and all unformalized analytic and numerical statements.

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